

Spartanflow

LinkedIn Content Engine

Built in 4 hours with Claude Code.

Voice-matched. AI-powered. \$3/month to run.

4-5 hrs

Build Time

\$3-4/mo

Running Cost

6

Workflows

30+

Auto Topics

A step-by-step walkthrough of how I built a self-hosted LinkedIn content engine that generates posts in my voice, creates branded images, and manages everything through a clean dashboard. No agency. No templates. Just automation.



Hari

CEO & Co-Founder, SpartanFlow

spartanflow.com

What I Built

A self-hosted web app that handles the entire LinkedIn content pipeline: topic discovery, post generation in my voice, AI images, and a dashboard to review posts.

1 Voice-Matched Writing

72-question voice interview turned into a system prompt. A humanizer strips AI patterns. Posts sound like me, not a chatbot.

2 AI Image Generation

Gemini generates branded images relevant to each post. My headshot and the SpartanFlow logo get composited automatically.

3 Smart Topic Discovery

25+ topics from my real experience building SpartanFlow, plus trending AI topics from Hacker News and RSS feeds.

4 Rating & Learning

I rate each post 1-5. I log LinkedIn engagement metrics. The AI reads past ratings and learns what works.

5 Custom Brain Dumps

I type raw notes about something I built. The engine turns messy thoughts into a polished LinkedIn post in two minutes.

6 Slack Notifications

Daily ping: 'Hey, 3 new posts ready for review.' Open dashboard, approve, copy, paste to LinkedIn. Under 5 minutes.

How It's Built

FRONTEND

Next.js 14 + React + Tailwind CSS

Dark-themed dashboard with post cards, filter tabs, calendar view, and inline editing.

DATABASE

SQLite via Prisma ORM

Lightweight, file-based. Stores posts, ratings, engagement metrics, and settings.

POST GENERATION

Claude API (Sonnet 4.6)

Voice profile system prompt + category templates + humanizer. ~\$0.02 per post.

IMAGE GENERATION

Gemini 2.5 Flash

Content-aware image prompts. Headshot + logo composited with Sharp. ~\$0.04/image.

IMAGE PROCESSING

Sharp (Node.js)

Headshot circle crop, logo overlay, gradient footer, branded composition.

TOPIC DISCOVERY

Hacker News API + RSS Feeds

Free. Pulls from 8+ curated AI/tech feeds plus 25+ personal experience topics.

NOTIFICATIONS

Slack Incoming Webhooks

Block Kit messages: 'Hey, 3 posts ready.' Links back to the dashboard.

HOSTING

Self-hosted / Vercel free tier

Runs anywhere. SQLite file is the database. No external services beyond AI APIs.

Total Monthly Cost: \$3-4

~30 posts/month: Claude API ~\$0.60 | Gemini images ~\$1.20 | Hosting: \$0
Topic discovery, RSS feeds, Slack webhooks: all free. Less than a coffee.

5 Phases, 4 Hours

Each phase had verification steps. Claude Code had to prove it worked before moving on.

0

PHASE 0 ~10 min

Project Scaffold

Next.js app, Prisma schema, all 30+ files created. Dev server running, health check passing, zero build errors.

1

PHASE 1 ~7 min

Voice Engine + Humanizer

3,481-character voice profile from a 72-question interview. Humanizer checks 12 AI patterns. 29/29 tests passing.

2

PHASE 2 ~15 min

Image Generation Pipeline

Gemini API integration, Sharp compositing, headshot circle crop, logo overlay. Content-aware prompts per topic.

3

PHASE 3 ~30 min

Dashboard UI

Dark theme, PostCards with copy/download/approve, filter tabs, calendar view, mobile responsive. 6 sample posts.

4

PHASE 4 ~60 min

Full Pipeline + Polish

End-to-end generation, topic discovery, Slack notifications, settings page, rating system, brain dump feature.

KEY INSIGHT

Verification gates at every phase meant nothing broke silently. Claude Code had to prove each phase worked before moving on. That discipline is what made 4 hours enough.

5 Minutes Per Day

- 1 Slack notification: '3 new posts ready for review'
- 2 Open dashboard. Glance at posts and images.
- 3 Quick-edit anything that feels off.
- 4 Hit 'Approve' on the ones I like.
- 5 Copy text + download image.
- 6 Open LinkedIn. Paste. Attach image. Post.
- 7 Mark as posted. Log engagement later.

Want something like this for your business?

SpartanFlow builds AI-powered outbound systems that fill your pipeline.

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